Week 1

Introduction

The global transition towards electric vehicles (EVs) represents a monumental shift in transportation, promising a future characterized by reduced emissions and enhanced sustainability. However, the widespread adoption and success of this paradigm shift are inextricably linked to the development of a robust, efficient, and accessible charging infrastructure. This inaugural week of lectures provides a foundational understanding of the intricate world of EV charging systems, dissecting their core components, operational principles, and the underlying technologies that facilitate the safe and effective transfer of energy to EV batteries. From the fundamental distinctions between AC and DC charging to the complex communication protocols that govern vehicle-charger interactions, this week aims to equip learners with essential knowledge in this rapidly evolving field.

Understanding EV charging extends far beyond the simple act of plugging in a vehicle; it involves a sophisticated interplay of power electronics, communication networks, and advanced battery management systems. As EVs become increasingly prevalent, the demand for diverse and accessible charging solutions intensifies, necessitating a deep dive into topics such as different charging modes, the challenges of battery health monitoring, and the various types of charging connectors and cables. This week's content sets the stage by outlining the course's scope, exploring the subsystems of an EV, detailing key battery characteristics, and introducing the diverse methods and modes of battery charging, all of which are crucial for designing and implementing the next generation of EV charging solutions.

Core Concepts

Term	Definition	Significance
`EV Charging System`	The complete infrastructure and technology used to replenish the energy in an electric vehicle's battery.	Essential for the operation and widespread adoption of electric vehicles, enabling energy replenishment.
`Battery Management System (BMS)`	An electronic system that monitors and manages a battery pack, ensuring its safe operation, optimal performance, and longevity.	Dictates the charging profile and communicates critical battery data to the charger, ensuring safe and efficient charging.

Term	Definition	Significance
`EVSE`	Electric Vehicle Supply Equipment, which includes the charging station, cables, and connectors that deliver electricity to the EV.	The interface between the power grid and the electric vehicle, ensuring safe and controlled power delivery.
`Onboard Charger (OBC)`	The power conversion unit located inside the electric vehicle, responsible for converting AC input to DC for the battery.	Enables AC charging by performing the necessary power conversion within the vehicle, making it a key component for AC charging.
`Off-board Charger`	A term referring to the DC EVSE itself, where the entire power conversion process takes place outside the vehicle.	Highlights the key architectural difference of DC charging, enabling higher power delivery and faster charging times.
`CCCV Charging`	A charging mode that applies a constant current (CC) until the battery voltage reaches a maximum, then switches to constant voltage (CV) until the current tapers off.	The most common and effective charging strategy for lithium-ion batteries, balancing fast charging with battery health and safety.
`PLC`	Power Line Communication, a method of communication that uses existing electrical wiring to transmit data.	Enables high-level communication between EV and EVSE, especially for DC charging, by sharing data over power lines.
`CCS`	Combined Charging System, a type of EV charging connector that supports both AC and DC charging through a single vehicle inlet.	Reduces inventory for EV manufacturers and offers flexibility for users, widely adopted globally.
`Unity Power Factor`	A condition where the current drawn from the AC source is in	Crucial for efficient power transfer, reducing losses,

Term	Definition	Significance
	phase with the voltage, minimizing reactive power.	and optimizing the sizing of power components in the charging system.
`THD (Total Harmonic Distortion)`	A measure of the harmonic content present in a waveform, indicating how much the waveform deviates from a pure sine wave.	Low THD in current drawn from the grid prevents pollution of the electrical network and ensures grid stability.

Detailed Analysis

Introduction

The initial lecture, **Introduction to the EV Charging System**, provided a comprehensive overview of the EV charging ecosystem. It began by detailing the fundamental building blocks of both AC and DC chargers, emphasizing the critical role of the power conversion unit. This unit, comprising AC-DC and isolated DC-DC converters, transforms electrical power to meet the battery's specific charging requirements. Communication aspects were highlighted as crucial for safe battery charging, particularly for DC chargers where external power conversion necessitates robust data exchange. The course scope was outlined, covering various charging procedures and protocols, including AC Type 2 and DC standards like CCS2 and Bharat EVC DC001 (DC001).

An Electric Vehicle (EV) was presented as a complex system with several interconnected subsystems:
* **Mechanical Subsystem**: Includes the gearbox, differential, and linkages.
* **Propulsion Subsystem**: Comprises the electric motor (e.g., PMSM, induction) and a power electronics converter that converts battery DC to motor AC. A vehicle controller (ECU) manages this converter.
* **Energy Storage Subsystem (Accumulator)**: Typically a battery pack, managed by the Battery Management System (BMS).
* **Accumulator Management Unit (BMS)**: Ensures safe operation, cell balancing, temperature monitoring, and dictates charging profiles.
* **Vehicle Communication**: Both intra-vehicle and inter-vehicle communication.

Key EV battery information was discussed, noting variations in capacity and nominal voltage across vehicle segments:
* **Electric Two-Wheelers (E-2W)**: 1.2-8 kWh, 48-72V.
* **Electric Three-Wheelers (E-3W)**: 3.6-8 kWh, 48-72V.

* **Electric Cars (E-car)**:
* Low-Voltage (LV) Powertrain: ~72V, ~21 kWh (older models).
* High-Voltage (HV) Powertrain: 350-500V, 30-80 kWh (modern cars), requiring distinct DC charging systems.
Chargers must accommodate voltage ranges from 75% to 115% of nominal.

Various EV battery charging methods were introduced:
* **Conductive Charging**: Most common, physical connection via cables and plugs.
* **Contactless Charging**: Inductive or capacitive, wireless energy transfer, typically static. Dynamic contactless charging allows charging while in motion.
* **Battery Swapping**: Rapid replacement of a discharged battery with a charged one.

Battery charging modes were detailed, emphasizing the **CCCV (Constant Current followed by Constant Voltage) Mode** as the most common for lithium-ion batteries:
* **CC Mode**: Constant current, risk of overheating.
* **CV Mode**: Constant voltage, risk of high inrush currents.
* **CCCV Mode**: Combines CC (for fast charging) and CV (for full charge), often with a pre-charge stage.
* Other modes include CPCV, multi-step CCCV, and pulse charging.

The necessity of a dedicated EV charger was underscored due to the need for charging profile management (dictated by BMS), power conversion (AC-to-DC), protection and safety features, and communication/monitoring capabilities. The building blocks of an EV charger include the power conversion unit (AC-DC, isolated DC-DC, filters), communication ports (Proximity Pilot, Control Pilot, CAN, PLC), the EVSE itself, and compatible connectors/cables.

2. Building Blocks of an AC EV Charger

The lecture on **Building Blocks of an AC EV Charger** focused on the architecture where the primary power conversion occurs inside the vehicle.
* **General Building Blocks**: Any EV charger includes a Power Conversion Unit (AC-DC, isolated DC-DC, filters), Communication Ports (proximity pilot, control pilot, CAN), EVSE, and Connectors/Cables.
* **AC EVSE Architecture**:
* Receives single or three-phase AC input from the grid (2 kW to 22 kW).
* Its output is also AC, supplied directly to the vehicle.
* Contains safety interlocks, a controller, and sensors for metering and protection.
* Crucially, the AC EVSE does not perform any power conversion. Its role is smart switching, monitoring, and safety.
* **Onboard Charger (OBC)**:

* Located inside the vehicle.
* Consists of a rectifier (AC to DC) and an isolated DC-DC converter (for controlled DC output and safety isolation).
* Protective devices are also present.
* The final DC output is fed to the EV battery.
* **Role of BMS**: The BMS is the "master" controller, monitoring battery parameters and dictating the optimal charging profile. It communicates requirements to the OBC controller, which adjusts power electronic switches. The BMS also communicates with the AC EVSE controller to manage the charging session.
* **Communication Ecosystem**: Involves internal communication (BMS ↔ OBC Controller) and external communication (BMS ↔ AC EVSE Controller, AC EVSE Controller ↔ CMS).

3. Building Blocks of an DC EV Charger

The lecture **Building Blocks of an DC EV Charger** contrasted AC charging by detailing the "off-board" nature of DC charging.
* **DC EVSE Architecture**:
* Performs the entire power conversion process outside the vehicle.
* Designed for high-power applications (15 kW to 400 kW), typically drawing three-phase AC input.
* **Rectifier**: Converts incoming AC to DC.
* **Isolated DC-DC Converter**: Provides electrical isolation and regulates DC voltage/current to match BMS requirements.
* Includes safety interlocks and protective devices.
* The regulated DC output is supplied directly to the EV battery, bypassing the vehicle's onboard charger.
* **Role of BMS and Communication**:
* BMS remains the master, determining the charging profile.
* The DC EVSE has a dedicated Supply Equipment Communication Controller (SECC) which acts as the slave.
* High-level communication (e.g., CAN, PLC) is established between the BMS and SECC for continuous data exchange.
* The SECC controls the rectifier and isolated DC-DC converter based on BMS instructions.
* SECC also communicates with a Central Management System (CMS) for monitoring and metering.
* **Key Differences and Advantages**:
* **Power Conversion Location**: AC = onboard; DC = off-board.
* **Power Levels**: DC chargers offer significantly higher power for "fast charging."
* **Vehicle Weight**: DC charging reduces vehicle weight by moving heavy conversion

components off-board.
* **Use Cases**: AC for overnight/extended; DC for quick top-ups.
* **Complexity**: DC EVSEs are more complex due to housing the entire power conversion unit.
* **Standards and Interoperability**: Adherence to standards is crucial for seamless operation across different EVSEs and EVs.

4. Review of EV Charger Types and Nomenclatures

The lecture **Review of EV Charger Types and Nomenclatures** provided a comprehensive look at the classifications, connectors, and communication protocols.
* **Public Charging Station Architecture**: Comprises multiple EVSEs (AC or DC), connected to the grid, and managed by a Central Management System (CMS) via protocols like OCPP. High-level communication (PLC or CAN) between EVSE and EV's BMS is essential for DC charging.
* **Modes of EV Charging**:
* **Mode 1 (AC)**: Simplest, direct connection to standard socket, onboard charger converts AC to DC.
* **Mode 2 (AC)**: Standard socket with in-cable control unit for safety.
* **Mode 3 (AC)**: Dedicated AC EVSE, controlled and safer, onboard charger converts AC to DC.
* **Mode 4 (DC)**: Dedicated DC EVSE, power conversion happens within EVSE, delivers DC directly to battery.
* **Levels of EV Charging (Indian Context)**:
* **Level 1 (AC)**: 230V, up to 3.5 kW.
* **Level 1 (DC)**: >= 48V to 72V, up to 15 kW.
* **Level 2 (AC)**: 380-400V, up to 22 kW.
* **Level 3 (AC)**: 200-1000V, > 22 kW.
* **Level 3 (DC)**: 200-1000V, up to 400 kW.
* **AC Charger Plugs and Vehicle Inlet Connectors**:
* **AC Type 1 (SAE J1772)**: US, Japan; single/dual-phase, PWM communication.
* **AC Type 2 (Mennekes)**: Europe, UK, India; single/three-phase, PWM communication.
* **GB/T AC**: China; single/three-phase, CP/CC communication.
* **AC001 (India)**: IEC 60309 industrial plug, single-phase up to 3.3 kW, no dedicated EV-EVSE communication.
* **DC Charger Plugs and Vehicle Inlet Connectors**:
* **CHAdeMO**: Japan; DC only, CAN communication.
* **CCS1**: US, Japan; combines Type 1 AC with DC, PLC for DC, PWM for AC.
* **CCS2**: Europe, India; combines Type 2 AC with DC, PLC for DC, PWM for AC.
* **GB/T DC**: China; DC only, CAN communication.

* **BEVC-DC001 (DC001)**: India; for 48V/72V systems up to 15 kW, CAN communication, uses GB/T 20234.3 connector.
* **DC Connector for LEVs (Type 6, India)**: BIS standard for low-voltage LEVs, up to 12 kW, CAN communication.
* **Combined AC and DC Connector for LEVs (Type 7, India)**: BIS standard for LEVs, up to 12 kW DC and 7 kW AC, CAN communication.
* **Regional and Indian Standards**: Highlighted the global preferences (US: Type 1/CCS1; Europe/India: Type 2/CCS2; China: GB/T) and India's complex landscape supporting both high-voltage (CCS2, Type 2) and low-voltage (AC001, DC001, Type 6, Type 7) powertrains with various communication standards (ISO 15118, CAN).

5. Revisiting Diode Bridge Rectifier

The lecture **Revisiting Diode Bridge Rectifier** introduced the fundamental AC-DC conversion stage.
* **EV Charging Ecosystem and Power Conversion Units**: Reemphasized the two main battery voltage categories (low: 48-72V; high: 350-500V) and the placement of the power conversion unit (off-board for DC EVSE, on-board for AC EVSE).
* **AC-DC Converter Requirements**:
* Convert AC to controlled and regulated DC output.
* Draw current with **Unity Power Factor**.
* Draw sinusoidal or nearly sinusoidal current.
* Ensure low **Total Harmonic Distortion (THD)** to prevent grid pollution.
* **Single-Phase Half-Bridge Diode Rectifier**:
* Described as an "uncontrolled rectifier" due to its inability to regulate output voltage.
* **Circuit**: Single AC source, one diode, resistive load.
* **Operation**: Diode conducts only during the positive half-cycle of the input AC voltage, producing a half-sine wave output voltage and current.
* **Average Output Voltage**: `Vs,peak / pi`.
* **Current Drawn**: Half-sine wave, identical to load current when conducting.
* **Diode Selection Criteria**:
* **Voltage Ratings**: Peak Repetitive Reverse Voltage (VRRM) must be at least `Vs,peak`.
* **Current Ratings**: Average Forward Current, Repetitive/Non-Repetitive Peak Forward Current, and `I square t rating` (for fuse selection) are critical.

Key Takeaways

- [ ] EV charging systems are essential for the widespread adoption of electric vehicles, involving complex power electronics and communication.

- [ ] The Battery Management System (BMS) is the "master" controller, dictating charging profiles and ensuring battery safety and longevity.
- [ ] AC charging utilizes an Onboard Charger (OBC) within the vehicle for AC-to-DC conversion, while DC charging uses an Off-board Charger (DC EVSE) for faster, higher-power conversion outside the vehicle.
- [ ] The CCCV (Constant Current followed by Constant Voltage) mode is the standard charging strategy for lithium-ion batteries, balancing speed and battery health.
- [ ] EV charging is classified by modes (Mode 1-4) and power levels, with Mode 4 and Level 3 DC representing fast charging.
- [ ] Robust communication protocols (e.g., PLC for CCS, CAN for CHAdeMO/GB/T/Indian standards) are critical for safe and efficient DC charging between the EVSE and the EV's BMS.
- [ ] AC-DC converters in EV chargers must provide controlled DC output, draw current with unity power factor, and exhibit low Total Harmonic Distortion (THD) to ensure grid compatibility.
- [ ] Various regional and national connector standards (e.g., Type 1, Type 2, CCS1, CCS2, CHAdeMO, GB/T, Indian standards like DC001, Type 6, Type 7) exist, necessitating interoperability.

Conclusion

This introductory week has laid a crucial groundwork for understanding the multifaceted world of EV charging systems. We've explored the fundamental subsystems of an electric vehicle, delved into the critical characteristics of EV batteries, and distinguished between the architectural nuances of AC and DC charging. The necessity of dedicated EV chargers, driven by the need for intelligent power conversion, safety protocols, and precise battery management, has been clearly established. From the basic principles of diode rectification to the sophisticated communication networks that enable seamless energy transfer, the complexity and ingenuity inherent in modern EV charging infrastructure are evident.

As the electric vehicle market continues its rapid expansion, the evolution of charging infrastructure will remain paramount. The principles discussed this week—from the importance of unity power factor and low THD for grid integration to the intricacies of various charging modes and connector standards—will serve as essential building blocks for future innovations. This foundational understanding is vital for anyone involved in designing, deploying, or maintaining the robust, user-friendly, and sustainable charging solutions that will power the next generation of electric mobility and accelerate the global transition towards a cleaner transportation future.